CLOUDASTICK SYSTEMS — SALESFORCE PARTNER

ABC Pharmacy Salesforce Solution

Technical Documentation & Architecture Decision Record

PROJECT
ABC Pharmacy Order Management System

API VERSION
67.0 — Winter '26

PREPARED BY
Abdelrhman Ehab Abdelkhalek

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ASSESSMENT
Salesforce Entry — Cloudastick Systems

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1. Solution Overview

ABC Pharmacy required a Salesforce environment to manage its product catalog, customer registry, and order lifecycle — integrated with an existing **Ionic web application** used by end-users for shopping, and a **warehouse interface** used by fulfilment staff to manage outbound deliveries.

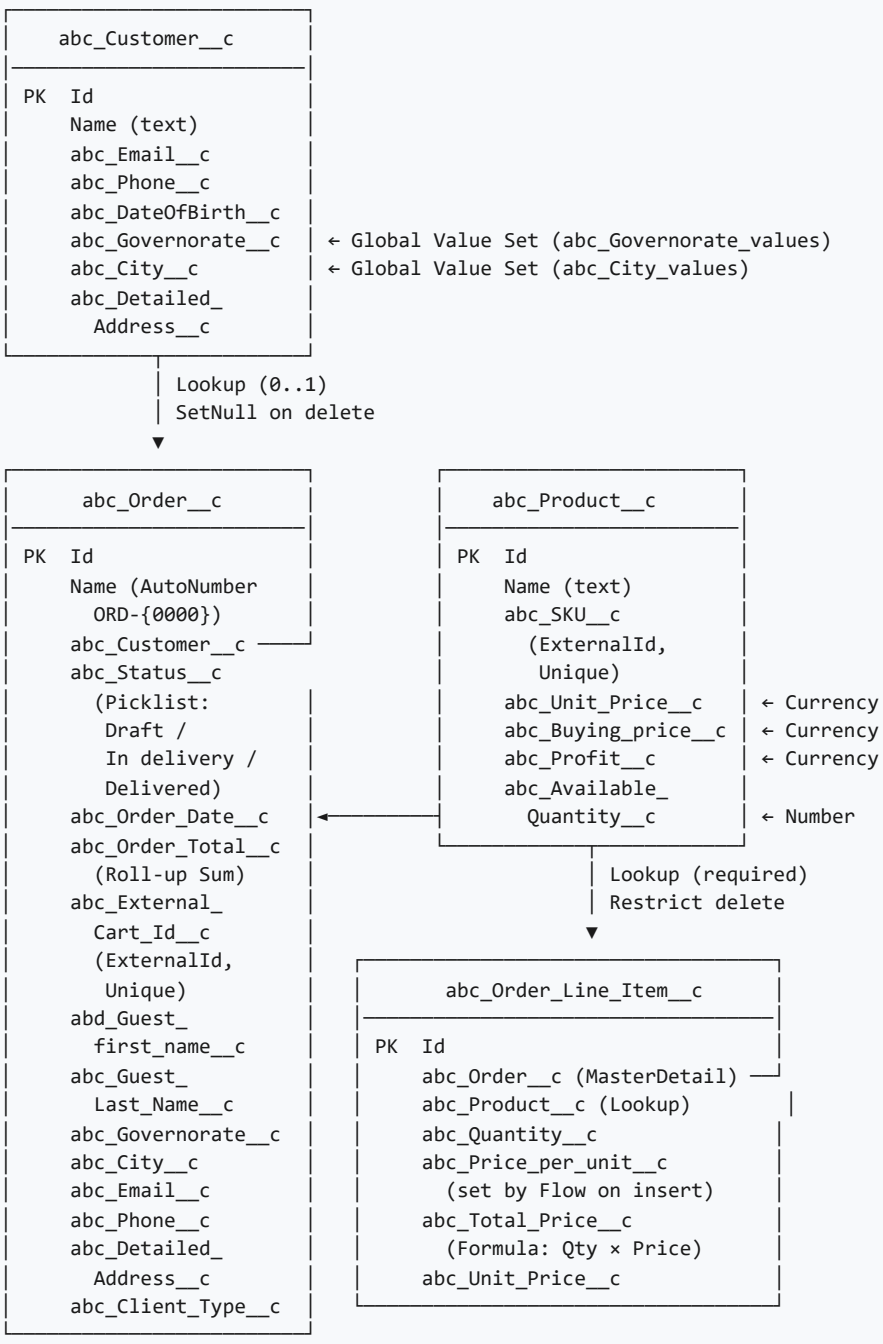
Architecture at a Glance

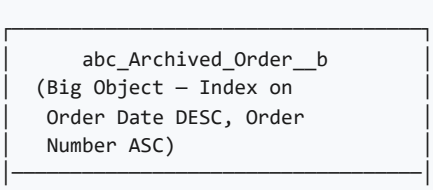
Layer	Component	Audience
Data	5 Custom Objects + 1 Big Object + 1 Log Object	All users
Automation	Record-Triggered Flows (price freeze, stock deduction), Roll-up Summary (order total)	Internal
Integration	REST API — GET /products, POST /cart	Ionic web app
UI — Internal	Lightning App + Custom Lightning Record Pages	Admin, Pharmacist
UI — Warehouse	Visualforce Page (abc_WarehouseOrders)	Warehouse user
Archival	Batch Apex → Big Object	Scheduled / Admin-triggered
Observability	abc_API_Log__c (structured error records)	Admin
Security	Custom Profiles (pharmacist, warehouse) + FLS	All users

> **Convention:** All custom metadata follows the `abc_` API name prefix as required, ensuring clear namespace separation from standard Salesforce objects and future managed packages.

2. Data Model & Design Decisions

2.1 Entity Relationship Diagram





```
│     abc_Order_Date__c (DateTime)  │
│     abc_Order_Number__c (Text)    │
│     abc_Status__c (Text)          │
│     abc_Order_Total__c (Number)   │
│     abc_Customer_Email__c         │
│     abc_External_Cart_Id__c       │
│     abc_Governorate__c (Text)     │
│     abc_City__c (Text)            │
│     abc_Client_Type__c (Text)     │
│     abc_Detailed_Address__c       │
└───────────────────────────────────┘


┌───────────────────────────────────┐
│         abc_API_Log__c            │
├───────────────────────────────────┤
│     Name (AutoNumber LOG-{0000})  │
│     abc_Class_Name__c             │
│     abc_Method_Name__c            │
│     abc_Error_Message__c          │
│     abc_Stack_Trace__c            │
└───────────────────────────────────┘
```

2.2 Objects Reference

Object	Type	Purpose
`abc_Customer__c`	Custom Object	Registered pharmacy customer profiles with address, contact, and demographics
`abc_Product__c`	Custom Object	Pharmacy product catalog with pricing (unit/buying/profit) and stock levels
`abc_Order__c`	Custom Object	Pharmacy orders supporting both registered and guest users via the Ionic app
`abc_Order_Line_Item__c`	Custom Object	Individual ordered items within an order (Master-Detail → Order, Lookup → Product)
`abc_Archived_Order__b`	Big Object	Immutable long-term archive of delivered orders older than 1 year
`abc_API_Log__c`	Custom Object	Structured, queryable error log written by REST API classes on exception

2.3 Key Design Decisions & Rationale

Decision 1 — Custom Objects vs. Standard Salesforce Objects

DECISION

Built five custom objects with the `abc_` prefix rather than repurposing standard objects (Contact → Customer, Product2 → Product, Opportunity → Order).

ALTERNATIVE CONSIDERED

Using standard objects (Contact, Product2, Opportunity + OpportunityLineItem) would have required fewer object licences and some built-in features like Pricebooks.

WHY THIS CHOICE

Standard objects carry implicit business meaning (e.g. Opportunity implies a sales pipeline, not a completed order), complex lifecycle rules, and fields irrelevant to pharmacy operations. Custom objects give a clean domain model that maps directly to pharmacy concepts, avoids polluting the standard CRM with unrelated records, and makes permissions simpler — we grant access only to the objects this business needs.

Decision 2 — Nullable Lookup vs. Required Master-Detail for Customer → Order

DECISION

`abc_Order__c.abc_Customer__c` is a nullable Lookup (SetNull on delete) rather than a required Master-Detail or a required Lookup.

ALTERNATIVE CONSIDERED

A required Lookup or Master-Detail would enforce that every order belongs to a customer. This is simpler but breaks the guest checkout use case.

WHY THIS CHOICE

The Ionic app supports **guest checkout** — users can place orders without registering. Forcing a Customer record would require creating a ghost customer for every guest, leading to data pollution. Instead, guest orders store contact details directly on the Order (`abd_Guest_first_name__c` , `abc_Guest_Last_Name__c` , `abc_Email__c`). SetNull on customer delete preserves order history without cascading deletions.

TRADE-OFF ACCEPTED

Guest and registered orders require conditional logic when displaying customer details. This is handled in both the Visualforce controller and the API — an acceptable complexity given the business requirement for guest checkout.

Decision 3 — Master-Detail vs. Lookup for Line Item → Order

DECISION

`abc_Order_Line_Item__c.abc_Order__c` is a Master-Detail relationship to `abc_Order__c`.

WHY THIS CHOICE

Master-Detail enables three critical features at zero extra code cost: (1) **Roll-up Summary** on the Order for the real-time total, (2) **cascade delete** so that deleting an order automatically deletes all its line items, and (3) enforced referential integrity — a line item cannot exist without a parent order. A Lookup would require Apex triggers to replicate these behaviours.

Decision 4 — Before-Save Flow vs. Apex Trigger for Price Freezing

DECISION

Price freezing (copying `Product.abc_Unit_Price__c` → `LineItem.abc_Price_per_unit__c`) is handled by a before-save Record-Triggered Flow.

Approach	Transaction Cost	Admin Maintainable	Retroactive Change Risk
✓ Before-save Flow	No extra DML	Yes ✓	None (frozen on create)
After-save Apex Trigger	Extra DML (update)	No	None
Formula Field	None	Yes	High — always reflects current price
Default Field Value	None	Yes	None — but can't reference Product field

WHY THIS CHOICE

A before-save Flow runs in the *same transaction* as the insert, requires no additional DML statement (unlike an after-save trigger that would need an update), and can be modified by a Salesforce admin without code deployment. A formula field was rejected because it would always show the *current* product price — meaning historical orders would retroactively reflect price changes, which is incorrect for financial records.

Decision 5 — Big Object vs. Custom Object for Archiving

DECISION

Delivered orders older than 1 year are archived to `abc_Archived_Order__b` (a Big Object) rather than to a standard custom object.

Approach	Storage Cost	Queryable	Scalability
✓ Big Object	Included (no limit)	Via SOQL with index	Petabyte-scale
Custom Object	Counts against 10GB default	Full SOQL	Limited by storage plan
External database	External cost	Via callout only	Unlimited

WHY THIS CHOICE

A pharmacy accumulating years of order history will generate hundreds of thousands of records. Standard custom object storage (10GB default) would eventually be exhausted, requiring costly storage upgrades. Big Objects are designed precisely for this use case: high-volume, append-only, historically queryable data at no additional storage cost. The compound index (`Order Date DESC` , `Order Number ASC`) ensures efficient retrieval of recent archives.

Decision 6 — Custom REST API vs. Standard Salesforce API for Ionic Integration

DECISION

Two custom `@RestResource` Apex classes rather than exposing the Ionic app to the standard Salesforce REST/SOAP API directly.

WHY THIS CHOICE

The standard Salesforce REST API would require the Ionic app to: (1) know internal Salesforce record IDs (not SKU codes), (2) make multiple round-trips to create an order and its line items, (3) implement idempotency logic itself, and (4) understand Salesforce-specific error structures. A custom REST API abstracts all of this: it accepts SKU codes, creates both Order and Line Items in a single atomic request, handles idempotency via `externalCartId`, and returns clean business-friendly JSON responses with standard HTTP status codes (201, 400, 404, 422, 500).

Decision 7 — without sharing on WarehouseOrdersController

DECISION

`abc_WarehouseOrdersController` is declared `without sharing`.

WHY THIS CHOICE

The warehouse user needs to see *all* orders regardless of record ownership. Orders are created by the Ionic API (running as a connected app user) or by pharmacists, not by the warehouse user. With `with sharing`, the warehouse profile would only see orders they own — which would be none. Using `without sharing` is appropriate here because access to this specific page is already controlled by profile-level page access permissions (only the warehouse profile can access `/apex/abc_WarehouseOrders`).

Decision 8 — abc_API_Log__c vs. Platform Events vs. Debug Logs

DECISION

API errors are persisted to `abc_API_Log__c` records (custom object).

Approach	Persistent?	Admin Visible?	Queryable?	Complexity
✓ abc_API_Log__c (Custom Object)	Yes	Yes	Yes (SOQL)	Low
System.debug (Debug Logs)	Until purged (2MB limit)	Dev only	No	None
Platform Events	72h (standard)	Needs subscriber	Via subscription	Medium
External monitoring (e.g. Datadog)	Yes	External only	Yes	High

WHY THIS CHOICE

Debug logs require an active debug session to capture and are capped at 2MB — they are a development tool, not a production monitoring solution. Platform Events require a separate subscriber process. A custom object is the simplest approach that gives admins a permanent, queryable record of all API failures directly in Salesforce — no additional tooling required.

3. Apex Components

3.1 abc_ProductAPI — Product List Endpoint

Type	`@RestResource · global with sharing`
URL	`/services/apexrest/abc/products`
Method	GET
Purpose	Returns all Pharmacy Products with `abc_Available_Quantity__c > 0`, sorted alphabetically. Filters out-of-stock items so the Ionic app never presents products users cannot order.
Error Handling	Full try/catch — exceptions are logged via `abc_APILogger` and returned as `{ "success": false, "error": "..." }` with HTTP 500.
Sharing	`with sharing` — respects FLS and object permissions of the calling user.

3.2 abc_CartAPI — Shopping Cart Endpoint

Type	`@RestResource · global with sharing`
URL	`/services/apexrest/abc/cart`
Method	POST
Purpose	Creates `abc_Order__c` + one `abc_Order_Line_Item__c` per item in a single atomic operation. Accepts Salesforce IDs or SKU codes as product references.
Idempotency	Checks `abc_External_Cart_Id__c` before inserting. Duplicate cart IDs return HTTP 200 with the existing order instead of creating duplicates.
Validation	HTTP 400 (empty items / blank productId / zero quantity / invalid customerId) · HTTP 404 (product not found) · HTTP 422 (stock insufficient)
Phone Handling	`abc_Phone__c` is stored as a Number field in the org. The API accepts the phone as a string, strips non-numeric characters, and converts before assignment.

3.3 abc_APILogger — Error Logger

Type	`public without sharing`

Purpose	Centralised error logger called by all API classes. Inserts an `abc_API_Log__c` record with class name, method, message, and full stack trace.
Resilience	Wraps its own DML in a try/catch and falls back to `System.debug` if the log insert fails — ensuring the API never crashes because of a logging failure.
Why without sharing	Logs must always be written regardless of the calling user's sharing access. Restricting by sharing rules could silently drop error records.

3.4 abc_ArchiveOrdersBatch — Archival Batch Class

Type	`public with sharing · implements Database.Batchable<SObject>`	
Query Filter	`abc_Status__c = 'Delivered' AND abc_Order_Date__c < TODAY - 1 year`	
Write Method	`Database.insertImmediate()`	— required for Big Object DML
Batch Size	200 (default, configurable at execution time)	
Email Logic	Prefers linked `abc_Customer__r.abc_Email__c`; falls back to the order-level `abc_Email__c` for guest orders.	
Note	Archives are copies — source records are not deleted. Deletion is a separate, explicit business decision.	
Run via	`Database.executeBatch(new abc_ArchiveOrdersBatch(), 200);`	

3.5 abc_WarehouseOrdersController — Visualforce Controller

Type	`public without sharing`
Purpose	Backs `abc_WarehouseOrders.page`. Loads all orders with their line items in a single SOQL query (sub-query on the Master-Detail relationship).
OrderWrapper	Inner class pairing each order record with a resolved icon (⏳ / ⏳ / 🚚 / ✅) and its list of line items. Status icons take priority over date icons.
Icon Logic	Delivered → ✅ · In delivery → 🚚 · Order Date = Today → ⏳ · Order Date < Today → ⏳
Row Expansion	Client-side JavaScript toggle — no server round-trip, instant UX.

3.6 Test Classes

Test Class	Covers	Scenarios Tested
`abc_ProductAPITest`	abc_ProductAPI	Returns only in-stock products; empty catalog returns 200 with count 0
`abc_CartAPITest`	abc_CartAPI	Success by SF ID; success by SKU; idempotency (200 on retry); 400 for empty items; 400 for blank productId; 400 for invalid customerId; 404 for unknown product; 422 for over-stock request

abc_ArchiveOrdersBatchTest	abc_ArchiveOrdersBatch	Old delivered orders are archived; recent delivered orders excluded; draft orders excluded; customer email preferred over order-level email
abc_WarehouseOrdersControllerTest	abc_WarehouseOrdersController	All 4 icon types independently tested; line items loaded correctly; all orders returned

4. Salesforce Features Used

Feature	Component	Why Chosen
Custom Lightning App	`abc_PharmaD`	Provides a branded navigation space for pharmacists and admins, surfacing only the pharmacy-relevant tabs (Products, Customers, Orders, Line Items). Keeps the pharmacy domain separate from standard Salesforce Sales/Service apps.
Custom Objects & Fields	6 objects, 30+ fields	Clean domain model with pharmacy-specific semantics. Field types chosen deliberately: AutoNumber for Order Name (human-readable IDs), Roll-up Summary for totals (no Apex needed), Formula for line item totals, Currency for prices.
Big Object	`abc_Archived_Order__b`	Designed for unlimited-scale append-only historical data. Prevents storage exhaustion as order history grows year over year. `Database.insertImmediate()` writes bypass standard DML governor limits.
Record-Triggered Flows (before-save & after-save)	`abc_product_and_line_item_price_match`	Freezes unit price at order time without extra DML (before-save runs in the same transaction). Admin-maintainable without code deployments. Chosen over Apex trigger for lower maintenance burden.
Roll-Up Summary Field	`abc_Order_Total__c`	Real-time, always-accurate order total with zero Apex code. Only possible because Line Item has a Master-Detail to Order — this was a deliberate data model choice to unlock this free feature.

Feature	Component	Why Chosen
REST Apex (@RestResource)	abc_ProductAPI, abc_CartAPI	Required for Ionic app integration. Custom REST preferred over standard object API because the cart creation requires multi-step atomic logic (stock validation + parent/child DML + idempotency) that a single standard API call cannot express.
Batch Apex (Database.Batchable)	`abc_ArchiveOrdersBatch`	Standard Salesforce pattern for large-scale, scheduled data processing. Processes records in configurable chunks (200 default) to stay within governor limits. Cannot be replaced by a Scheduled Flow because Flows cannot call `Database.insertImmediate()` for Big Objects.
Visualforce Page	`abc_WarehouseOrders`	Specifically required by the assessment. Server-side icon resolution in Apex controller (clean separation of logic from markup) combined with client-side JavaScript for row expansion (instant UX, no server round-trip).
Custom Profiles	pharmacist, warehouse	Principle of least privilege. Warehouse users get read-only order/product access and can only view the Visualforce page. Pharmacist users manage the full order lifecycle. Role separation mirrors real pharmacy operations.
Lightning Record Pages (Flexipages)	4 custom record pages	Activated on all custom objects to provide a polished Lightning Experience UI beyond the default page layout. Allows drag-and-drop component arrangement without code.
Global Value Sets	abc_Governorate_values, abc_City_values	Egyptian governorates and cities are shared across `abc_Customer__c` and `abc_Order__c`. Centralising in Global Value Sets means adding a new city requires one update, not one per object. Ensures consistency across all picklists referencing the same geography.

Feature	Component	Why Chosen
External ID Fields	abc_SKU__c, abc_External_Cart_Id__c	SKU lets the Ionic app reference products by real-world codes instead of opaque Salesforce IDs. External Cart ID enables idempotency — the API checks this field before inserting so retried requests do not create duplicate orders. Both fields are indexed for efficient lookups.
Custom WebLinks (Buttons)	abc_Warehouse_Orders_Page, abc_Warehouse_Orders_List	Allows the warehouse user to navigate to the Visualforce page directly from the Order record detail page and from the Orders list view — without needing to know the page URL.
Error Logging Object	`abc_API_Log__c`	Provides a persistent, admin-visible, SOQL-queryable audit trail of API failures. Chosen over debug logs (transient, dev-only) and Platform Events (requires a subscriber process).
Custom Notification Type	`abc_low_stock_alert`	Provides real-time, in-app and mobile alerting to warehouse staff the moment a product's stock drops critically low — see Section 7.3.

5. API Reference

Authentication

All endpoints require a valid Salesforce OAuth2 Bearer token.

```
Authorization: Bearer <access_token>
```

For production server-to-server integration, the **Connected App + JWT Bearer Flow** is recommended. For testing, the Username-Password flow works with a Connected App.

5.1 GET /services/apexrest/abc/products

Returns all pharmacy products with available stock (`abc_Available_Quantity__c > 0`), sorted alphabetically by name.

Request

```
GET https://<instance-url>/services/apexrest/abc/products
Authorization: Bearer <token>
```

Response 200 — Success

```
{
  "success": true,
  "count": 2,
  "data": [
    {
      "id": "a01bm000001XxxxAAA",
      "name": "Amoxicillin 250mg",
      "sku": "SKU-AMOX-250",
      "unitPrice": 35.0,
      "availableQuantity": 100
    },
    {
      "id": "a01bm000001XyyyAAA",
      "name": "Paracetamol 500mg",
      "sku": "SKU-PARA-500",
      "unitPrice": 15.0,
      "availableQuantity": 200
    }
  ]
}
```

Response 500 — Error

```
{ "success": false, "error": "<exception message>" }
```

5.2 POST /services/apexrest/abc/cart

Creates a pharmacy order with one or more line items atomically.

Request

```
POST https://<instance-url>/services/apexrest/abc/cart
Authorization: Bearer <token>
Content-Type: application/json

{
  "externalCartId":  "CART-IONIC-001",
  "customerId":      "003bm000001XxxxAAA",
  "guestFirstName":  "Ahmed",
  "guestLastName":   "Ali",
  "governorate":     "Cairo",
  "city":            "Nasr City",
  "email":           "ahmed@example.com",
  "phone":           "01001234567",
  "detailedAddress": "123 Tahrir Street, Apt 5",
  "items": [
    { "productId": "SKU-PARA-500", "quantity": 2 },
    { "productId": "a01bm000001XxxxAAA", "quantity": 1 }
  ]
}
```

Request Field Reference

Field	Type	Required	Notes
`externalCartId`	String	No	Enables idempotency — safe to re-send on retry
`customerId`	String	No	Salesforce ID of abc_Customer__c; omit for guest orders
`guestFirstName`	String	No	Used when no customerId is provided
`guestLastName`	String	No	
`governorate`	String	No	Must match abc_Governorate_values picklist

Field	Type	Required	Notes
`city`	String	No	Must match abc_City_values picklist
`email`	String	No	Order contact email
`phone`	String	No	Numeric digits — non-numeric characters are stripped automatically
`detailedAddress`	String	No	Free text address
`items`	Array	Yes	Minimum 1 item
`items[].productId`	String	Yes	Salesforce record ID or abc_SKU__c value
`items[].quantity`	Integer	Yes	Must be > 0

Response Codes

Status	Condition	Body
201 Created	Order created successfully	`{ "success": true, "orderId": "...", "orderNumber": "ORD-0001" }`
200 OK	Idempotent — cart already exists	`{ "success": true, "message": "Cart already exists.", "orderId": "...", "orderNumber": "..." }`
400 Bad Request	Validation failure (empty items, blank productId, zero quantity, invalid customerId)	`{ "success": false, "error": "..." }`
404 Not Found	Product not found by ID or SKU	`{ "success": false, "error": "Product not found: SKU-X" }`
422 Unprocessable	Requested quantity exceeds available stock	`{ "success": false, "error": "Insufficient stock..." }`
500 Server Error	Unexpected exception (logged to abc_API_Log__c)	`{ "success": false, "error": "..." }`

6. Org Access

Instance URL

```
https://orgfarm-7e74891727-dev-ed.develop.my.salesforce.com
```

Reviewer Account (System Administrator)

Username	`assessments@cloudastick.com.pharmd.com`
Email	`assessments@cloudastick.com`
Profile	System Administrator
License	Salesforce

> **Note:** Use the "Forgot Password" flow at the instance URL above to set the password for this account before logging in.

Warehouse User

Profile	warehouse
Visualforce Page	`/apex/abc_WarehouseOrders`
Access via Order record	"Warehouse Orders View" button on any Pharmacy Order record
Access via list view	"Warehouse Orders View" in the Orders list view actions dropdown

Postman Collection

The file `ABC_Pharmacy_Postman_Collection.json` is included in the project root. Import it into Postman, run the **Auth → Get Access Token** request first, then test both API endpoints from the Products and Cart folders.

7. Automation, Validation Rules & Custom Notifications

This section catalogs the declarative automation layer of the org — the Record-Triggered Flows, Validation Rules, and Custom Notification driving business logic that sits outside the Apex components described in Section 3.

7.1 Record-Triggered Flows

Flow	Object	Trigger Timing	Purpose
`abc_product_and_line_item_price_match`	`abc_Order_Line_Item__c`	Before-Save, on Create	Freezes the line item's unit price from the linked product at creation time, so historical orders are unaffected by later product price changes. Full rationale in Decision 4, Section 2.3.
`abc_deduct_product_quantity_on_delivery`	`abc_Order__c`	After-Save, on Update	Deducts each line item's quantity from the matching product's Available Quantity the moment an order's status becomes "In delivery," mimicking stock leaving the warehouse, and alerts the warehouse team when a product's remaining stock drops to 1 or less.

Flow Detail — abc_product_and_line_item_price_match

Trigger	`abc_Order_Line_Item__c`	**Timing**	Before-Save · Create
Entry Filter	`abc_Product__c` is not null		
Action	Assigns `$Record.abc_Price_per_unit__c` = `$Record.abc_Product__r.abc_Unit_Price__c`		

Flow Detail — abc_deduct_product_quantity_on_delivery

Trigger	`abc_Order__c`	**Timing**	After-Save · Update
Entry Filter	`abc_Status__c` = 'In delivery', only when this is a new transition into that value		

DECISION

Entry criteria uses the native "Only when a record is updated to meet the condition requirements" trigger option (`doesRequireRecordChangedToMeetCriteria`) rather than a manual decision element comparing against `$Record__Prior` .

WHY THIS CHOICE

This is the platform-native way to express "fire only on this transition." It prevents the flow from re-running (and re-deducting stock) every time an order that is already "In delivery" is edited for an unrelated reason, without an extra visible decision node cluttering the canvas.

WHY AFTER-SAVE, NOT BEFORE-SAVE

Before-save (fast field update) flows may only write to the triggering record itself. This flow must update a different object — `abc_Product__c` — so an after-save context is architecturally required, not merely a style preference.

Logic Walkthrough

#	Element	Description
1	Get Records	All `abc_Order_Line_Item__c` for the order.
2	Get Records	The `abc_low_stock_alert` Custom Notification Type, and every active User on the `warehouse` profile (via a Profile lookup, then `User.ProfileId`) — fetched once per order, not per line item.
3	Loop	For each line item: get its product, subtract the line item quantity from `abc_Available_Quantity__c` , and queue the product into a collection.
4	Decision	If the product's new quantity is ≤ 1, call **Send Custom Notification** to the warehouse recipient list.
5	Update Records	Single bulk DML for every product touched by the order, run once after the loop — not once per iteration.

7.2 Validation Rules

Five active validation rules enforce data integrity across three custom objects, guarding against invalid pricing, quantities, dates, and incomplete guest/registered customer data.

Object	Rule	Condition	Error Message
`abc_Product__c`	`abc_price_can_not_be_0_or_less`	Unit Price or Buying Price is blank or ≤ 0	"the unit price or buying price can not be blank or 0 or less"
`abc_Order_Line_Item__c`	`abc_quantity_can_not_be_0`	`abc_Quantity__c` ≤ 0	"quantity can not be zero"
`abc_Order__c`	`Order_date_can_not_be_in_the_past`	`abc_Order_Date__c` < TODAY()	"Order date can not be in the past"
`abc_Order__c`	`abc_if_registered_then_fill_customer`	Client Type = "registered" AND `abc_Customer__c` is blank	"customer field must be filled when type value is registered"
`abc_Order__c`	`abc_if_guest_then_fill_first_last_name`	Client Type = "guest" AND (First Name or Last Name is blank)	"must fill first name and last name if type is guest"

WHY THIS CHOICE

Declarative validation rules keep these business constraints admin-maintainable and visible in Setup, without requiring an Apex trigger for checks that don't need cross-object logic — consistent with the "flow/declarative first" approach used for price freezing (Decision 4, Section 2.3).

7.3 Custom Notifications

Notification Type	`abc_low_stock_alert`
Label	Low Stock Alert
Fired By	`abc_deduct_product_quantity_on_delivery`
Trigger Condition	A product's `abc_Available_Quantity__c` is ≤ 1 immediately after a delivery-triggered deduction
Recipients	Every active User with the `warehouse` profile, resolved dynamically at run time — never a hardcoded User Id
Click-Through	Deep-links to the affected `abc_Product__c` record
Channels	Desktop and Mobile enabled

Message Template

```
{ProductName} is down to {AvailableQuantity} unit(s) in stock. Reorder soon.
```

> **WHY THIS CHOICE**
>
> A Custom Notification appears instantly in the Salesforce bell icon and on mobile for every warehouse user, with no email-relay configuration, and its `targetId` deep-links directly to the specific product — something a declarative workflow email alert cannot do. Recipients are resolved by profile membership at run time (Profile → User lookup) rather than a hardcoded User Id, so the alert keeps working if warehouse staffing changes.